Speech-Recognition

Exercises on N-gram Language Model, not part of LN but may be in the exam

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Abstract

Calculations on 3-grams and smoothing, Python program to handle bigrams

Prerequisites

Read through the slides spr-xx-language-model.pdf. For more details see the Jurafsky book.

Part 01 Trigrams

The chain rule of probability for a trigram results in:

$P(X_1 \cdots X_3) = P(X_1) \times P(X_2 \mid X_1) \times P(X_3 \mid X_{1:2})$

Write out the equation for trigram probability estimation by applying this (modifying Eq. 3.11, p. 32 in the book, slide x in PDF).

Next, write out all the non-zero trigram probabilities for the "I am Sam corpus":

Part 02 Probability of the sentence "i want chinese food"

Calculate the probability of the sentence "i want chinese food". Give two probabilities, one using Fig. 3.2 and the "useful probabilities" just below it on page 43, and another using the add-1 smoothed table on slide 21 (Fig. 3.7 of book). Assume the additional add-1 smoothed probabilities

$P(i \mid < s >) = 0.19$

and

$P(< /s > \mid food) = 0.40$

Part 03 Comparison of Probabilities

Which of the two probabilities you computed in the previous exercise is higher, unsmoothed or smoothed? Explain why.

Part 04 Bigrams of Corpus

We are given the following corpus, modified from the one in the chapter:

```
<s> I am Sam </s>
<s> Sam I am </s>
<s> I am Sam </s>
<s> I do not like green eggs and Sam </s>
```

Write a Python program with the following requirements:

- The sentences shall be implemented as strings.
- Words in a sentence are separated with space.
- Create a list of the given sentences.
- Include <s> and </s> in your counts just like any other word.
- Write a function that returns the relationship between the occurrence a specific bigram and all bigrams (apply a bigram language model).
- Write another function that returns the result with add-one smooting.
- Apply the function on P('Sam' | 'am')!